

df field theory

Vincent Wang

February 2026

1 Introduction

If $\text{char } K = p$, then $p \cdot k = 0$ for all $k \in K$.

Proof: I claim $n \cdot k = k \cdot (n \cdot 1)$ for all $n \in \mathbb{Z}_{\geq 0}$. For $n = 0$, we have $0 \cdot k = 0$ and $k \cdot (0 \cdot 1) = k \cdot 0 = 0$. Sps the result for n . Then

$$(n+1) \cdot k = n \cdot k + k = k \cdot (n \cdot 1) + k = k \cdot (n \cdot 1 + 1) = k \cdot ((n+1) \cdot 1),$$

completing the induction.

It follows $p \cdot k = k \cdot (p \cdot 1) = k \cdot 0 = 0$.

Let K be a field and let $p \in K[x]$ be nonzero and nonconstant. I claim that p is irred iff $\langle p \rangle$ is a prime ideal.

Sps p is irred and $p \mid q_1 q_2$. Consider $(p, q_1) = (r_1)$. Then $r_1 \in K^\times$ or is off by a unit from p . If $r_1 \in K^\times$, then $1 \in (p, q_1)$, so there exist $a, b \in K[x]$ such that $1 = pa + q_1 b \rightarrow q_2 = paq_2 + q_1 q_2 b \rightarrow p \mid q_2$. Else, $p \mid q_1$, so (p) is prime ideal.

Suppose (p) is prime. Suppose $p = q_1 q_2$ for nonconstant q_1, q_2 . Then $q_1 q_2 \in (p)$. Suppose $q_1 \in (p)$. Write $q_1 = pr_1$, so $pr_1 q_2 = p \rightarrow r_1 q_2 = 1$, absurd. Similarly, $q_2 \notin (p)$, so p is irred.

Claim: Let $P \leq R$ be a nonzero prime ideal of the PID R . Then P is maximal.

Proof: Sps FTSOC $P < Q < R$. Write $P = (p)$ for $p \neq 0$ and $Q = (q)$ for $q \notin R^\times$. We have $q \mid p$, so $p = qa$ for some $a \in R$. P is prime, so $p \mid q$ or $p \mid a$. If $p \mid q$, then $q = pb$, so $pq = pqab \rightarrow ab = 1$ (since $p, q \neq 0$), so $a, b \in R^\times$, so $(p) = (q)$, absurd.

Hence, $p \mid a$. Write $a = pa_1$, so $qa_1 = 1 \rightarrow a_1, q \in R^\times$, so $(q) = R$, absurd.

Let R be a commutative ring with identity $1 \neq 0$. I claim $(r) = R$ iff $r \in R^\times$.

Sps $(r) = R$. Then there exists $s \in R$ such that $rs = 1$, so $r \in R^\times$. Conversely, if $r \in R^\times$, then for all $a \in R$, we have $r(r^{-1}a) = a \in (r)$, so $(r) = R$.

Hence, we have $K[x]/(p)$ is a field.

Let K/F and let $\alpha \in K$ be algebraic over F . Then there exists a unique irreducible monic polynomial $m_{\alpha,F} \in F[x]$ such that α is a root of $m_{\alpha,F}$.

Proof: Consider the ring homomorphism $\varphi : F[x] \rightarrow K$ by $f(x) \mapsto f(\alpha)$. Since α is algebraic, $\ker \varphi \neq 0$. Since $F[x]$ is a PID, let $\ker \varphi = (m_{\alpha,F})$.

I claim that $m_{\alpha,F}$ is irred. Sps $m_{\alpha,F} = q_1 q_2$ for nonconstant q_1, q_2 . Then α is a root of at least one of q_1, q_2 . WLOG $q_1(\alpha) = 0$, so $q_1 \in \ker \varphi$. Write $q_1 = m_{\alpha,F} q_3$, so $m_{\alpha,F} = m_{\alpha,F} q_2 q_3$. Since $m_{\alpha,F} \neq 0$, we have $q_2 q_3 = 1 \rightarrow q_2, q_3 \in K^\times$, absurd.

Hence, $m_{\alpha,F}$ is irred.

Suppose $m'_{\alpha,F}$ is monic irred with α as a root. Then $m'_{\alpha,F} \in \ker \varphi$, so $m'_{\alpha,F} = q m_{\alpha,F}$. Since $m'_{\alpha,F}$ is irred, we have $q \in K^\times$. Since $m'_{\alpha,F}, m_{\alpha,F}$ are monic, we have $q = 1$, so $m_{\alpha,F}$ is unique.

Sps $K/L/F$ and α is algebraic over F (and therefore L). Then $m_{\alpha,L} \mid m_{\alpha,F}$ in $L[x]$.

Proof: α is a root of $m_{\alpha,F}$, which can be viewed as a polynomial of $L[x]$, so $m_{\alpha,L} \mid m_{\alpha,F}$.

Sps F/K is generated by a finite number of algebraic elements $\alpha_1, \alpha_2, \dots, \alpha_n$, so $F = K[\alpha_1, \alpha_2, \dots, \alpha_n]$. Define $F_i = K[\alpha_1, \alpha_2, \dots, \alpha_i]$, so $F_0 = K$ and $F_n = F$. Note that

$$[F : K] = [F_n : F_0] = \prod_{i=0}^{n-1} [F_{i+1} : F_i] \leq \prod_{i=0}^{n-1} d_{i+1},$$

where d_i is the degree of α_i over F (for $i \in [n]$), so F/K is finite.

Conversely, sps F/K is finite. Then F/K is algebraic. If $K = F$, we're done. Else, pick $\alpha \in F - K$. Let $L = K[\alpha]$. Then we have

$$[F : K] = [F : L][L : K].$$

Since $[L : K] > 1$, we have $[F : L] < [F : K]$, so by induction there exist algebraic $\alpha_1, \alpha_2, \dots, \alpha_n$ over K such that $F = K[\alpha_1, \alpha_2, \dots, \alpha_n]$.

Let K/F . Sps $\alpha, \beta \in K$ are algebraic over F . Then $F[\alpha, \beta]$ is finite and

therefore algebraic. It follows $\alpha \pm \beta, \alpha\beta, \alpha/\beta$ (for $\beta \neq 0$) are algebraic.

Let L/F be an extension. Then the set K of elements in L algebraic over F form a subfield of L .

Proof: $K \neq 0$ (since $F \leq K$) is closed under subtraction and division.

Sps $L/K/F$. Then L/F is algebraic iff L/K and K/F are algebraic.

Proof: Sps L/F is algebraic. Then $L/K, K/F$ are algebraic.

Sps $L/K, K/F$ are algebraic. Sps $\alpha \in L$. Write $p = \sum_{i=0}^n k_i x^i$, where $p \neq 0$ has α as a root, $k_i \in K$. Let $M = F(\alpha_0, \alpha_1, \dots, \alpha_n)$. We have $[M(\alpha) : M] < \infty$. Since α_i is algebraic over F for all $i \in [n]$, M/F is finite. Hence, $M(\alpha)/F$ is finite. Since $F(\alpha) \leq M(\alpha)$, α is algebraic over F .

Claim: If $K_1, K_2/F$ have bases $\{\alpha_i\}_{i \in [n]}, \{\beta_j\}_{j \in [m]}$, then

$$K_1 K_2 = F(\alpha_1, \alpha_2, \dots, \alpha_n, \beta_1, \beta_2, \dots, \beta_m).$$

Proof: Let $L = F(\alpha_i, \beta_j)$. Note that $K_1, K_2 \leq L$, so $K_1 K_2 \leq L$. Conversely, $\alpha_i, \beta_j \in K_1 K_2$ and $F \leq K_1 K_2$, so $L \leq K_1 K_2$. Hence, $K_1 K_2 = L$, as desired.

Claim: $K_1 K_2 = \langle k_1 k_2 : k_1 \in K_1, k_2 \in K_2 \rangle$.

Proof: Let $L = \langle k_1 k_2 : k_1 \in K_1, k_2 \in K_2 \rangle$, so $L \neq \emptyset$ is closed under subtraction. L is closed under multiplication.

Define $i : \mathbb{Q}[x]/(x^3 - 2) \rightarrow \mathbb{Q}(i)[x]/(x^3 - 2)$ by inclusion. Sps FTSOC $x^3 - 2$ is reducible in $\mathbb{Q}(i)$, so $\pi : \mathbb{Q}(i)[x]/(x^3 - 2) \rightarrow \mathbb{Q}(i)$ by $f(x) \mapsto f(\alpha)$ for some $\alpha \in \mathbb{Q}(i)$ such that $\alpha^3 - 2 = 0$ is an isomorphism.

Consider the (nonzero) field homomorphism $\varphi = \pi i : \mathbb{Q}[x]/(x^3 - 2) \rightarrow \mathbb{Q}(i)$. We have the field inclusions $\mathbb{Q} \hookrightarrow \text{im } \varphi \rightarrow \mathbb{Q}(i)$, so $[\text{im } \varphi : \mathbb{Q}] \mid [\mathbb{Q}(i) : \mathbb{Q}]$, or $3 \mid 2$, absurd.

$F(\{\alpha_i\})$ contains F and α_i for all $i \in [n]$, so $F(\{\alpha_i\}) \geq F(\alpha_i)$, so $F(\{\alpha_i\})$ contains the composite of the $F(\alpha_i)$.

Conversely, the composite of the $F(\alpha_i)$ contains F and all of the α_i , and therefore $F(\alpha_1, \alpha_2, \dots, \alpha_n)$.

If $[F(\alpha) : F]$ is odd, then $F(\alpha) = F(\alpha^2)$.

Proof: Since $\alpha^2 \in F(\alpha)$, we have $F(\alpha^2) \leq F(\alpha)$. Let $m_{\alpha, F}$ be odd degree. Write $m_{\alpha, F} = x \cdot P + Q$, where P, Q are even polynomials. Since $m_{\alpha, F}$ is odd degree, $P \neq 0$. Then $\alpha P(\alpha^2) + Q(\alpha^2) = 0$. By minimality of $m_{\alpha, F}$, we have $P(\alpha^2) \neq 0$,

since $P(x^2)$ has degree less than $m_{\alpha, F}$. Then $\alpha = -Q(\alpha^2)/P(\alpha^2) \in F(\alpha^2)$, so $F(\alpha) \leq F(\alpha^2)$.

Let $K_1, K_2 \leq K$ be such that $K_1, K_2/F$ are finite. Then $[K_1 K_2 : F] = [K_1 : F][K_2 : F]$ iff $K_1 \otimes_F K_2$ is a field.

Sps $K_1 \otimes_F K_2$ is a field. Let $K_1 = F[\alpha_1, \alpha_2, \dots, \alpha_n]$ and $K_2 = F[\beta_1, \beta_2, \dots, \beta_m]$. Define $\varphi : K_1 \otimes_F K_2 \rightarrow K_1 K_2$ by $\varphi(k_1 \otimes k_2) = k_1 k_2$.

Claim: Let $\varphi : K \rightarrow F(\alpha_1, \alpha_2, \dots, \alpha_n)$ be a field homomorphism such that $\alpha_i \in \text{im } \varphi$ for all $i \in [n]$. Then φ is surjective.

Define $\varphi : K_1 \times K_2 \rightarrow K_1 K_2$ by $\varphi(k_1, k_2) = k_1 k_2$. φ descends to the map $\Phi : K_1 \otimes_F K_2 \rightarrow K_1 K_2$.

$$\begin{aligned} \Phi((k_1 \otimes k_2)(k'_1 \otimes k'_2)) &= \Phi(k_1 k'_1 \otimes k_2 k'_2) = k_1 k'_1 k_2 k'_2 = (k_1 k_2)(k'_1 k'_2) \\ &= \Phi(k_1 \otimes k_2) \Phi(k'_1 \otimes k'_2), \end{aligned}$$

$\Phi(1 \otimes 1) = 1$, so Φ is a field homomorphism.

Surjective field homomorphism is field isomorphism, so $K_1 \otimes_F K_2 \cong K_1 K_2$.

Then $[K_1 K_2 : F] = [K_1 \otimes_F K_2 : F] = [K_1 : F][K_2 : F]$.

Sps $\varphi : R \rightarrow S$ is a ring isomorphism and R is a field. Then S is a field.

Proof: Since R is commutative, so is S . Sps $s_1 \in S - 0$. Then $s_1 = \varphi(r_1)$ for some $r_1 \neq 0$. Since R is a field, there exists $r_2 \in R$ such that $r_1 r_2 = 1$. Let $s_2 = \varphi(r_2)$. Then $s_1 s_2 = \varphi(r_1) \varphi(r_2) = \varphi(r_1 r_2) = \varphi(1) = 1$, so S is a field.

Define $\Phi : K_1 \otimes_F K_2 \rightarrow K_1 K_2$ as before. Φ as a linear map of F -vector spaces is bijective. Φ is a ring homomorphism, so Φ is a field isomorphism, so $K_1 \otimes_F K_2$ is a field.

Sps $p \in F[x]$ is cubic reducible with root $\alpha \in F$. Define $\varphi : F[x]/(p) \rightarrow F$ by $f(x) \mapsto f(\alpha)$. I claim that φ is an injective ring homomorphism.

$(a + bi)^3 = 2$. $a^3 - 3ab^2 = 2$ and $3a^2b - b^3 = 0$. If $b = 0$, then $a^3 = 2$, absurd. Else, $3a^2 = b^2$, absurd (since $\sqrt[3]{2}, \sqrt{2} \notin \mathbb{Q}$).

2 Splitting Fields and Algebraic Closures

Let $f \in F[x]$. Then there exists $K \geq F$ such that K is a splitting field of f .

Proof: I claim there exists $K \geq F$ such that f splits over K . If f splits over F , we're done. Else, pick $\alpha \notin F$ such that $f(\alpha) = 0$. Let $K = F(\alpha)$, so the number of unsplit factors of f strictly decreases. By induction, the result follows.

If K has no subfields for which f splits, we're done. Else, pick $K' < K$ for which f splits. Since $\dim K' < \dim K$ and there cannot exist an infinite descending chain of positive integers, there exists a splitting field of f .

Sps $[K : F] = 1$. Let $K = \langle k_0 \rangle$ for some $k_0 \in K$. In particular, there exists $f \in F$ such that $fk_0 = 1$, so $k_0 = f^{-1} \in F$. It follows $K \leq F$, so $K = F$.

Sps $f = x^n + \sum_{i=0}^{n-1} a_i x^i$, where $v_p(a_0) = 1$ and $p^2 \mid a_i$ for $i \in [n-1]$. Then I claim f is irred.

Consider $\mathbb{Q}(\zeta_p)$, where p is prime. Consider $\Phi_p(x) = x^{p-1} + x^{p-2} + \cdots + x + 1$, which has ζ_p as a root. I claim that Φ_p is irred.

Note that $\Phi_p(x) = \frac{x^p - 1}{x - 1}$ and $\Phi_p(x+1) = \frac{(x+1)^p - 1}{x} = \frac{1}{x} \sum_{i=1}^p \binom{p}{i} x^i = \sum_{i=1}^p \binom{p}{i} x^{i-1}$. Since $p \mid \binom{p}{i}$ for $i \in [p-1]$, we have that $\Phi_p(x+1)$ and therefore Φ_p is irred by Eisenstein.

$[\mathbb{Q}(\zeta_p, \sqrt[p]{2}) : \mathbb{Q}] = p(p-1)$ and $[\mathbb{Q}(\zeta_p) : \mathbb{Q}] = p-1$, so $[\mathbb{Q}(\zeta_p, \sqrt[p]{2}) : \mathbb{Q}(\zeta_p)] = p$. $\mathbb{Q}(\zeta_p, \sqrt[p]{2}) = \mathbb{Q}(\zeta_p)(\sqrt[p]{2})$.

Let K/F be an algebraic extension and let $F \leq R \leq K$ be a ring. Then R is a field.

Proof: R is closed under subtraction. Let $r \in R$ be nonzero. Since K/F is algebraic, there exists a minimal polynomial $p \in F[x]$ of r . Since $r \neq 0$, $p(0) \neq 0$. Let $p = \sum_{i=0}^n a_i x^i$ with $a_0 \neq 0$. Then in K , we have

$$r^{-1} = -a_0^{-1} \sum_{i=0}^{n-1} a_{i+1} r^i \in F(r) \leq R,$$

so R is closed under inverses. Since R is a ring, R is closed under multiplication. Hence, R is a subfield of K .

Let $\overline{F}$ be an algebraic closure of F . Then $\overline{F}$ is algebraically closed.

Proof: Let $f \in \overline{F}[x]$ and E be the splitting field of f . Sps $\overline{F} \neq E$, so pick $\alpha \in E - \overline{F}$ a root of f , so $\overline{F} < \overline{F}(\alpha)$. $\overline{F}(\alpha)/\overline{F}, \overline{F}/F$ are algebraic, so $\overline{F}(\alpha)/F$ is algebraic. Hence, there exists a nonzero polynomial $p \in F[x]$ such that $p(\alpha) = 0$. Then $\alpha \in \overline{F}$, absurd.

Sps $f(x, y)(x^2 + x + 1) + g(x, y)(y^3 - y^2 + 1) = 1$. Then take $x = \alpha, y = \beta$, α, β roots of their respective polynomials.

Sps $K_0 \leq K_1 \leq K_2 \leq \dots$ is an ascending chain of fields. Let $K = \bigcup_{n < \omega} K_n$. Then K has a field structure.

Sps $h \in K[x]$. Then $h \in K_n[x]$ for some $n \in \mathbb{Z}_{\geq 0}$. h has a root in $K_{n+1} \leq K$, so K is algebraically closed.

Let $K \geq F$ be an algebraically closed field. Then the elements $\bar{F} \leq K$ algebraic over F is an algebraic closure of F , and $\bar{F}$ is unique up to isomorphism.

$\bar{F}$ is subfield of K containing F . $\bar{F}$ is algebraic over F . Sps $f \in F[x]$. Then $f \in K[x]$, so f splits completely over K . Let $\alpha \in K$ be a root of f . $F(\alpha)$ is finite-dimensional, so α is algebraic over F , so $\alpha \in \bar{F}$.

Sps $F_1, F_2/F$. Consider $R = F_1 \otimes_F F_2$. Treat R, F_1, F_2 as vector spaces over F . We have (with Cardinal numbers) $[R : F] = [R_1 : F] \cdot [R_2 : F] \geq 1$, so $R \neq 0$. We have the inclusions $F_1 \rightarrow R$ by $x \mapsto x \otimes 1$ and $F_2 \rightarrow R$ by $y \mapsto 1 \otimes y$. Since $R \neq 0$ is commutative, there exists a maximal ideal $M \triangleleft R$, so the maps $F_1, F_2 \rightarrow R/M$ are field homomorphisms.

Let $F_1, F_2/F$ be algebraic closures of F . Let $F_3/F_1, F_2$ be a common extension and let F_4/F_3 be an algebraically closed extension. Then F_1, F_2 are equal to the set of algebraically closed elements F_5 over F in F_4 .

Let K/F be finite. Prove that K is a splitting field over F iff every irred in $F[x]$ that has a root in K splits completely in $K[x]$.

Sps K is splitting for $f \in F[x]$. Let $g \in F[x]$ be irred st $g(\alpha) = 0$ for some $\alpha \in K$.

Sps $\gcd(f, g) = 1$ in $F[x]$. Then $a(x)f(x) + b(x)g(x) = 1$ for $a, b \in F[x]$. Then $0 = a(\alpha)f(\alpha) + b(\alpha)g(\alpha) = 1$, absurd. Hence, $g \mid f$, so g splits completely.

Sps for every irred $g \in F[x]$ that has a root in K splits completely in $K[x]$. $K = F(\alpha_1, \alpha_2, \dots, \alpha_n)$, since K/F is finite. Let $g_i \in F[x]$ be the irred assoc to α_i . Note that $g = g_1 g_2 \cdots g_n$ splits completely over K .

Sps $K/K'/F$ st g splits over K' . Then the roots of g_i are in K' , so in particular $\alpha_i \in K'$, so $K \leq K'$, so $K = K'$.

Let K be a splitting field of $f \in F[x]$ and sps $g(\alpha) = 0$ for some $\alpha \in K$, where g is an irred. Consider the map $\varphi : F(\alpha) \rightarrow F(\alpha)$.

Sps $K \geq K_1, K_2 \geq F$ are splitting, say of $f_1, f_2 \in F[x]$. Then $K_1 K_2$ is splitting.

(a). I claim that $K_1 K_2$ is splitting for $f_1 f_2$. Since $K_1, K_2 \leq K_1 K_2$, $f_1 f_2$ splits. Conversely, if $K \geq K_3 \geq F$ is splitting for $f_1 f_2$, then K_3 is splitting for

f_1 , so $K_3 \geq K_1$. Similarly, $K_3 \geq K_2$, so $K_3 \geq K_1 K_2$.

(b). I claim that $K_1 \cap K_2$ is splitting for $\gcd(f_1, f_2)$. $\gcd(f_1, f_2)$ splits completely over K_1, K_2 , and therefore $K_1 \cap K_2$. Sps $\gcd(f_1, f_2)$ splits over K_3/F .

3 Separable and Inseparable Extensions

Let $f \in F[x]$ and E/F . Sps $\alpha \in E$ is a multiple root of f . Then α is a root of f' .

Proof: Write $f = (x - \alpha)^n g$ for $n > 1$ and $g \in E[x]$. Then

$$f' = n(x - \alpha)^{n-1}g + (x - \alpha)^n g' = (x - \alpha)^{n-1}(ng + (x - \alpha)g'),$$

so α is a root of f' .

Now suppose α is a root of f and f' . Sps FT SOC $f = (x - \alpha)g$ for $g(\alpha) \neq 0$ and $g \in E[x]$. Then $f' = g + (x - \alpha)g'$, so $f'(\alpha) = g(\alpha) \neq 0$, so α is not a root of f' .

Let $f = f_1 f_2 \cdots f_n \in F[x]$ for distinct monic irreducibles f_i and $\text{char } F = 0$. Let E be a splitting field of f . Each f_i is irreducible and therefore separable. Sps FT SOC f_i, f_j share a common root $\alpha \in E$. Then $m_{\alpha, F} \mid f_i, f_j$, so $f_i = f_j = m_{\alpha, F}$, absurd, so f is separable.

Conversely, sps f is separable. Write as a product of monic irreducibles $f = f_1 f_2 \cdots f_n$. Sps FT SOC $f_i = f_j$. Then f is clearly not separable, so the f_i are distinct.

Sps $(a_k x^k)' = k a_k x^{k-1} = 0$. Then $k a_k = 0 \rightarrow a_k = 0$ if $p \nmid k$. If $f' = 0$, write $f = \sum_{i=0}^m a_{pi} x^{pi}$.

Existence and Uniqueness of Finite Fields.

Existence. Let $f = x^{p^n} - x \in \mathbb{Z}_p[x]$ and E be the splitting field of f . Since f is separable, f has p^n distinct roots, say forming a set K . Note that in $E/\mathbb{Z}_p$, we have $(a - b)^{p^n} = a^{p^n} - b^{p^n} = a - b$ and $(ab^{-1})^{p^n} = ab^{-1}$ for $b \neq 0$, so K is a finite field of p^n elements.

Let K be a field of p^n elements and consider $f = x^{p^n} - x \in K[x]$. $K^\times$ is a multiplicative group of $p^n - 1$ elements, so by Lagrange $\alpha^{p^n-1} = 1$ for all $\alpha \in K$. Then for all $\alpha \in K$, we have $\alpha^{p^n} = \alpha$. Since f is degree p^n , f factors exactly as $f = \prod_{\alpha \in K} (x - \alpha)$. The roots of f are distinct and are equal to the elements of K , so K is splitting. The splitting field is unique up to isomorphism.

I claim that $x^d - 1 \mid x^n - 1$ iff $d \mid n$. If $d \mid n$, write $n = dq$, so $x^n \equiv (x^d)^q \equiv 1^q \equiv 1 \pmod{x^d - 1}$. Conversely, write $n = dq + r$ for $0 \leq r < d$. Then $x^n \equiv x^r \pmod{x^d - 1}$, so we have $x^d - 1 \mid x^r - 1$. Note $\deg(x^r - 1) \leq r < d$, so we have $r = 0$, ie $d \mid n$.

For a base $a > 1$, I claim that $a^d - 1 \mid a^n - 1$ iff $d \mid n$. The forward direction is the same as before. For the converse, consider $0 \leq a^r - 1 < a^d - 1$, so $r = 0$.

I claim that $x^{p^m} - x \mid x^{p^n} - x$ iff $m \mid n$. We have

$$x^{p^m} - x \mid x^{p^n} - x \iff x^{p^m-1} - 1 \mid x^{p^n-1} - 1 \iff p^m - 1 \mid p^n - 1 \iff m \mid n.$$

It follows we have the inclusion $\mathbb{F}_{p^m} \leq \mathbb{F}_{p^n}$ for $m \mid n$. Conversely, note that $[\mathbb{F}_{p^m} : \mathbb{F}_p] = m$ since $|\mathbb{F}_{p^m}| = p^m$, so by multiplicativity of extension degrees, we have $m \mid n$.

Let $f \in \mathbb{Z}_p[x]$ be irred of degree n . I claim that $f = \prod_{i=0}^{n-1} (x - \alpha^{p^i}) \in \mathbb{F}_{p^n}[x]$, where the α^{p^i} are distinct.

Let $\mathbb{F}_{p^m}$ be the splitting field of f . Write $f = \prod_{i=0}^{n-1} (x - \alpha_i)$. Let $\Phi : \mathbb{F}_{p^m}[x] \rightarrow \mathbb{F}_{p^m}[x]$ be the Frobenius automorphism extended to polynomials. Note that $f \in \mathbb{F}_p[x]$ is fixed by Φ , so

$$\prod_{i=0}^{n-1} (x - \alpha_i) = \prod_{i=0}^{n-1} (x - \alpha_i^p).$$

Sps FTSOC $\alpha_i^{p^j} = \alpha_i$ for some $0 < j < n$. Rename α_i to α , so $\alpha \in \mathbb{F}_{p^j} \rightarrow \mathbb{F}_p(\alpha) \leq \mathbb{F}_{p^j}$, so $\mathbb{F}_p(\alpha)$ has degree less than n over $\mathbb{F}_p$. On the other hand, $m_{\alpha, \mathbb{F}_p} = f$, so $\mathbb{F}_p(\alpha) \cong \mathbb{F}_p[x]/(f)$ has degree n over $\mathbb{F}_p$, absurd.

Then WLOG $\alpha_i = \alpha_{i+1}^p$ for $i \in [n-1]$. After renaming, we have $f = \prod_{i=0}^{n-1} (x - \alpha^i)$, as desired.

Let $f \in \mathbb{F}_p[x]$ be irred of degree n . I claim that $f \mid x^{p^n} - x$. Let α be a root of f , so $\mathbb{F}_p(\alpha) \cong \mathbb{F}_{p^n}$. f is separable and any root β of f is a root of $x^{p^n} - x$, so $f \mid x^{p^n} - x$, as desired.

For $d \mid n$, let $\mathcal{F}_d$ denote the set of irreducibles in $\mathbb{F}_p$ with degree $d \mid n$. The set of polynomials with degree d in $\mathbb{F}_p$ is finite, so $\mathcal{F}_d$ is finite. Hence, $\prod_{f \in \mathcal{F}_d} f = \prod \mathcal{F}_d$ is well-defined. We have $\prod \mathcal{F}_d \mid x^{p^d} - x$ and $x^{p^d} - x \mid x^{p^n} - x$, so $\prod \mathcal{F}_d \mid x^{p^n} - x$. It follows $\prod_{d \mid n} \prod \mathcal{F}_d \mid x^{p^n} - x$.

Conversely, suppose $f \mid x^{p^n} - x$ is an irreducible of degree d . Identify $x^{p^n} - x \in \mathbb{F}_{p^n}[x]$. Since $x^{p^n} - x$ splits over $\mathbb{F}_{p^n}$, f splits over $\mathbb{F}_{p^n}$. Let $\alpha \in \mathbb{F}_{p^n}$ be a root of

f . Then $\mathbb{F}_p(\alpha)$ has degree d over $\mathbb{F}_p$. Since $\mathbb{F}_{p^n}$ has degree n over $\mathbb{F}_p$, we have $d \mid n$.

Since $x^{p^n} - x$ is separable, $x^{p^n} - x$ factors as a product of distinct irreducibles over $\mathbb{F}_p$. Hence, $x^{p^n} - x \mid \prod_{d \mid n} \prod \mathcal{F}_p$, so $x^{p^n} - x = \prod_{d \mid n} \prod \mathcal{F}_p$.

Claim: For any prime p and $a \in \mathbb{F}_p^\times$, $f = x^p - x + a$ is irreducible and separable over $\mathbb{F}_p$.

Note that if α is a root of f , then so is $\alpha + 1$:

$$f(\alpha + 1) = (\alpha + 1)^p - (\alpha + 1) + a = \alpha^p - \alpha + a = f(\alpha) = 0.$$

Pick $E/\mathbb{F}_p$ so that f has a root α in E . Since $a \neq 0$, we have $\alpha \notin \mathbb{F}_p$.

f splits as $f = \prod_{i=0}^{n-1} (x - (\alpha + i))$. In particular, f is separable. Suppose FTSOC that $f = f_1 f_2$ for nontrivial $f_1, f_2 \in \mathbb{F}_p[x]$. Since the roots of f are distinct, the decomposition $f = f_1 f_2$ corresponds to a nontrivial partition of the roots $\{\alpha + i : i \in [p-1]\}$, ie a partition of $[p-1]$.

Let $[p-1] = S \cup T$, where $S, T \neq \emptyset$. Then the second leading coefficient of f_1 is $c = -(|S|\alpha + \sum_{k \in S} k)$. If $c \in \mathbb{F}_p$, then $|S|\alpha \in \mathbb{F}_p \rightarrow \alpha \in \mathbb{F}_p$ (since $0 < |S| < p$), so $\alpha \in \mathbb{F}_p$, absurd. Hence, f is irreducible.

We have $(1+x)^{p^n} = ((1+x)^n)^p = (1+x^p)^n$ in $\mathbb{F}_p$. Comparing coefficients of x^{pi} , we have $\binom{pn}{pi} = \binom{n}{i} \pmod{p}$.

4 Cyclotomic Polynomials

Let p, l be primes. We factorize $\phi_l(x) = \frac{x^l - 1}{x - 1}$ over $\mathbb{F}_p$.

If $p = l$, then $x^l - 1 = (x - 1)^l$, so $\phi_l(x) = (x - 1)^{l-1}$, since $\mathbb{F}_l[x]$ is an integral domain.

Suppose $p \neq l$. Let ζ be a primitive l th root of unity. There exists minimal $f \in \mathbb{Z}^+$ such that $p^f \equiv 1 \pmod{l}$. I claim the minimal $n \in \mathbb{Z}^+$ such that $\zeta \in \mathbb{F}_{p^n}$ is $n = f$.

Indeed, we have $x^l - 1 \mid x^{p^f} - x$ since $l \mid p^f - 1$. Conversely, if $x^l - 1 \mid x^{p^n} - x$, then $x^l - 1 \mid x^{p^n-1} - 1 \rightarrow l \mid p^n - 1 \rightarrow f \mid n$, so $n \geq f$.

Hence, $\phi_l(x)$ is a product of $(l-1)/f$ irreducibles of degree f . Since $x^l - 1$ is separable, so is $\phi_l(x) \mid x^l - 1$, so the irreducibles are distinct.

Let $p \in \mathbb{Z}[x]$ be monic nonconstant. I claim that the list $S = \{P(n) : n \in \mathbb{Z}^+\}$ contains infinitely many distinct prime factors.

Sps FTSOC S has finitely many primes p_i for $i \in [n]$. Let $P = \prod_{i \in [n]} p_i$. Since p is nonzero monic, there exists $N \in \mathbb{Z}^+$ such that $a = P(N) \neq 0$. Define $q(x) = a^{-1}p(N + aPx)$. Note that in $\mathbb{Z}_a$, we have $p(N + aPx) = p(N) = 0$, so $q \in \mathbb{Z}[x]$. Further, in $\mathbb{Z}_{aP}$, we have $p(N + aP(x+1)) - p(N + aPx) = 0$, so $P \mid q(x+1) - q(x)$. We compute $q(0) = 1$, so by induction $q(n) = 1$ in $\mathbb{Z}_P$ for all $n \in \mathbb{Z}^+$.

Since p is nonconstant and $aP \neq 0$, so is q . Hence, there exists $n \in \mathbb{Z}^+$ such that $|q(n)| > 1$. Then $q(n)$ contains a prime not in S , so $p(N + aPn)$ contains a prime not in S , absurd (note $N + aPn > 0$).

Claim: Let p be an odd prime. Sps $\phi_m(a) = 0$ in $\mathbb{Z}_p$ for some $m \in \mathbb{Z}^+$ and $a \in \mathbb{Z}$. Then $p \nmid a$ and $|a| = m$ in $\mathbb{Z}_p^\times$.

Proof: Note that $\phi_m(x) \mid x^m - 1$, so we have $p \mid a^m - 1$, so $a \perp p$ and $|a| \mid m$. Since $p \nmid m$, $x^m - 1$ is separable. From

$$x^m - 1 = \phi_m(x) \prod_{d \mid m, d < m} \phi_d(x),$$

we have that $\phi_d(a) \neq 0$ for all $d \mid m$ and $d < m$. It follows for all $m' \mid m$ such that $m' < m$ that $x^{m'} - 1 \neq 0$ from the above formula, since if $d \mid m'$ and $m' \mid m$, we have $d \mid m$, so $|a| = m$.

Let $a \in \mathbb{Z}$. If p is an odd prime with $p \mid \phi_m(a)$, then $p \mid m$ or $p \equiv 1 \pmod{m}$.

Proof: From the previous result, if $p \mid m$, we're done. Else, $p \nmid a$ and $|a| = m$. Since $a^{p-1} = 1$ by Fermat's Little theorem, we have $m \mid p - 1$.

Claim: Given $m \in \mathbb{Z}^+$, there are infinitely many primes p such that $p \equiv 1 \pmod{m}$.

Proof: Consider the set $S = \{\phi_m(a) : a \in \mathbb{Z}^+\}$. $\phi_m(x) \in \mathbb{Z}[x]$ is monic nonconstant, so by the first lemma, S contains infinitely many odd prime factors p . If $p \mid \phi_m(a)$, we have $p \mid m$ or $p \equiv 1 \pmod{m}$. Since m has finitely many prime factors, there are infinitely many primes with $p \equiv 1 \pmod{m}$.